

Module 4: Calculus

Instructional Hours: 4

1 Limits

1.1 Definition of Limits

The limit of a function $f(x)$ as x approaches a particular value a is the value that $f(x)$ gets closer to as x gets closer to a . Mathematically, we express the limit of a function $f(x)$ as $x \rightarrow a$ as:

$$\lim_{x \rightarrow a} f(x) = L$$

This means that as x approaches a , the function $f(x)$ approaches L .

1.1.1 Content Curated Video

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<https://www.youtube.com/embed/k1Q7xZ0PU5U>



Evaluating a limit by factoring

1.2 Evaluating Limits Using Substitution

The simplest technique for evaluating limits is to directly substitute the value of x into the function if the function is continuous at that point.

Example 1. Evaluate $\lim_{x \rightarrow 3} (2x + 5)$.

Solution:

$$\lim_{x \rightarrow 3} (2x + 5) = 2(3) + 5 = 6 + 5 = 11$$

1.3 Limits Involving Factoring

When substitution results in an indeterminate form (such as $\frac{0}{0}$), factoring the expression may help simplify the function to find the limit.

Example 2. Evaluate $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$.

Solution: First, factor the numerator:

$$x^2 - 4 = (x - 2)(x + 2)$$

Now, cancel the common factor $x - 2$:

$$\lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{x-2} = \lim_{x \rightarrow 2} (x+2)$$

Now, substitute $x = 2$:

$$2 + 2 = 4$$

$$\text{Thus, } \lim_{x \rightarrow 2} \frac{x^2-4}{x-2} = 4.$$

1.4 Limits Involving Rationalizing

For limits involving square roots or other roots, rationalizing the numerator or denominator can help simplify the expression.

Example 3. Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x+1}-1}{x}$.

Solution: Multiply the numerator and the denominator by the conjugate of the numerator:

$$\frac{\sqrt{x+1}-1}{x} \cdot \frac{\sqrt{x+1}+1}{\sqrt{x+1}+1} = \frac{(\sqrt{x+1}-1)(\sqrt{x+1}+1)}{x(\sqrt{x+1}+1)}$$

Simplify the numerator:

$$= \frac{(x+1)-1}{x(\sqrt{x+1}+1)} = \frac{x}{x(\sqrt{x+1}+1)}$$

Cancel the common factor x :

$$= \frac{1}{\sqrt{x+1}+1}$$

Now substitute $x = 0$:

$$= \frac{1}{\sqrt{0+1}+1} = \frac{1}{2}$$

$$\text{Thus, } \lim_{x \rightarrow 0} \frac{\sqrt{x+1}-1}{x} = \frac{1}{2}.$$

1.5 Limits at Infinity

When x approaches infinity, the behavior of the function can often be evaluated by simplifying the expression and considering the dominant terms.

Example 4. Evaluate $\lim_{x \rightarrow \infty} \frac{3x^2+5x+2}{2x^2-x+1}$.

Solution: For large x , the highest powers of x in the numerator and denominator dominate, so divide the numerator and denominator by x^2 :

$$\lim_{x \rightarrow \infty} \frac{3x^2+5x+2}{2x^2-x+1} = \lim_{x \rightarrow \infty} \frac{3 + \frac{5}{x} + \frac{2}{x^2}}{2 - \frac{1}{x} + \frac{1}{x^2}}$$

As $x \rightarrow \infty$, the terms with $\frac{1}{x}$ and $\frac{1}{x^2}$ approach 0 :

$$= \frac{3 + 0 + 0}{2 + 0 + 0} = \frac{3}{2}$$

Thus, $\lim_{x \rightarrow \infty} \frac{3x^2 + 5x + 2}{2x^2 - x + 1} = \frac{3}{2}$.

1.6 Advanced Technique: L'Hôpital's Rule

L'Hôpital's rule can be used to evaluate limits that result in indeterminate forms like $\frac{0}{0}$ or $\frac{\infty}{\infty}$. The rule states that:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} \quad \text{if the limit on the right exists.}$$

Example 5. Evaluate $\lim_{x \rightarrow 0} \frac{\sin(x)}{x}$.

Solution: This limit results in the indeterminate form $\frac{0}{0}$, so we apply L'Hôpital's rule. First, differentiate the numerator and the denominator:

$$f'(x) = \cos(x), \quad g'(x) = 1$$

Now, apply the rule:

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = \lim_{x \rightarrow 0} \frac{\cos(x)}{1} = \cos(0) = 1$$

Thus, $\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$.

2 Continuity

2.1 Content Curated Video

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<https://www.youtube.com/embed/ki8F-HBuNbg>



2.2 Definition of Continuity

A function $f(x)$ is said to be continuous at a point $x = c$ if the following conditions are met:

1. Defined: $f(c)$ is defined.
2. Limit Exists: The limit of $f(x)$ as x approaches c exists.

3. **Equality of Limit and Function Value:** The limit of $f(x)$ as x approaches c is equal to the function value $f(c)$, i.e.,

$$\lim_{x \rightarrow c} f(x) = f(c)$$

If $f(x)$ is continuous at every point in its domain, we say that $f(x)$ is continuous on its domain.

2.3 Types of Discontinuities

There are several types of discontinuities:

1. **Point Discontinuity (Removable Discontinuity):** Occurs when the limit of $f(x)$ exists as x approaches c , but $f(c)$ is either not defined or not equal to the limit e.g. for

$$f(x) = \frac{x^2 - 1}{x - 1}$$

at $x = 1$. The limit exists but $f(1)$ is not defined.

2. **Jump Discontinuity:** Occurs when the left-hand limit and the right-hand limit as x approaches c are different, e.g. for

$$f(x) = \begin{cases} x + 1 & \text{if } x < 0 \\ x - 1 & \text{if } x \geq 0 \end{cases}$$

at $x = 0$. The left-hand limit is 1, and the right-hand limit is -1.

3. **Infinite Discontinuity (Vertical Asymptote):** Occurs when the limit of $f(x)$ approaches infinity or negative infinity as x approaches c , e.g. for

$$f(x) = \frac{1}{x}$$

at $x = 0$. The limit does not exist because it approaches infinity.

4. **Oscillatory Discontinuity:** Occurs when the function oscillates between different values as x approaches c , and the limit does not settle to a single value, e.g. for

$$f(x) = \sin\left(\frac{1}{x}\right)$$

at $x = 0$. The function oscillates infinitely as x approaches 0.

2.4 Solving Problems of Continuity

To determine the continuity of a function at a point:

1. Check if the function is defined at the point.
2. Calculate the left-hand limit and right-hand limit as x approaches the point.
3. Compare these limits and the function value to determine if the function is continuous.

Example 6. Determine if the function $f(x) = \frac{x^2-4}{x-2}$ is continuous at $x = 2$.

Solution: First, we simplify $f(x)$:

$$f(x) = \frac{x^2 - 4}{x - 2} = \frac{(x - 2)(x + 2)}{x - 2}$$
$$f(x) = x + 2 \text{ for } x \neq 2$$

The function $f(x)$ simplifies to $x + 2$ for all $x \neq 2$. We need to check the limit as x approaches 2 :

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} (x + 2) = 4$$

However, $f(x)$ is not defined at $x = 2$. Thus, there is a removable discontinuity at $x = 2$.

Example 7. Analyze the continuity of the function

$$f(x) = \begin{cases} x^2 & \text{if } x < 1 \\ 2x + 1 & \text{if } x \geq 1 \end{cases}$$

at $x = 1$.

Solution: Calculate the left-hand limit:

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} x^2 = 1$$

Calculate the right-hand limit:

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (2x + 1) = 3$$

Since the left-hand limit (1) and right-hand limit (3) are different, $f(x)$ has a jump discontinuity at $x = 1$.

Example 8. Determine the continuity of the function

$$f(x) = \frac{x^2 \sin\left(\frac{1}{x}\right)}{x^2 + 1}$$

at $x = 0$.

Solution: First, check if $f(x)$ is defined at $x = 0$. The function is not explicitly defined at $x = 0$, so we consider the limit: Calculate the limit as x approaches 0 :

$$\lim_{x \rightarrow 0} \frac{x^2 \sin\left(\frac{1}{x}\right)}{x^2 + 1}$$

We know that $-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$, so:

$$-\frac{x^2}{x^2 + 1} \leq \frac{x^2 \sin\left(\frac{1}{x}\right)}{x^2 + 1} \leq \frac{x^2}{x^2 + 1}$$

Both $\frac{x^2}{x^2 + 1}$ and $-\frac{x^2}{x^2 + 1}$ approach 0 as $x \rightarrow 0$:

$$\lim_{x \rightarrow 0} \frac{x^2}{x^2 + 1} = 0$$

$$\lim_{x \rightarrow 0} -\frac{x^2}{x^2 + 1} = 0$$

By the Squeeze Theorem:

$$\lim_{x \rightarrow 0} \frac{x^2 \sin\left(\frac{1}{x}\right)}{x^2 + 1} = 0$$

Hence, $f(x)$ is continuous at $x = 0$ if we define $f(0) = 0$.

3 Differentiation

3.1 Differentiation from First Principles

To find the derivative of a function $f(x)$ from first principles, use the limit definition:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Example 9. Differentiate $f(x) = x^2 + 3x + 2$.

Solution:

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 + 3(x+h) + 2 - (x^2 + 3x + 2)}{h}$$

Expanding $f(x+h)$:

$$(x+h)^2 + 3(x+h) + 2 = x^2 + 2xh + h^2 + 3x + 3h + 2$$

Subtracting $f(x)$:

$$f(x+h) - f(x) = (x^2 + 2xh + h^2 + 3x + 3h + 2) - (x^2 + 3x + 2) = 2xh + h^2 + 3h$$

So:

$$f'(x) = \lim_{h \rightarrow 0} \frac{2xh + h^2 + 3h}{h} = \lim_{h \rightarrow 0} (2x + h + 3) = 2x + 3$$

3.2 Rules of Differentiation

3.3 Power Rule

For $f(x) = x^n$, the derivative is:

$$f'(x) = nx^{n-1}$$

Example 10. Differentiate $f(x) = 4x^3 - 5x + 1$.

Solution:

$$f'(x) = 12x^2 - 5$$

3.4 Product Rule

For $f(x) = u(x) \cdot v(x)$, the derivative is:

$$f'(x) = u'(x) \cdot v(x) + u(x) \cdot v'(x)$$

Example 11. Example Differentiate $f(x) = x^2 \sin(x)$.

Solution: Let $u(x) = x^2$ and $v(x) = \sin(x)$:

$$u'(x) = 2x, \quad v'(x) = \cos(x)$$

$$f'(x) = (2x) \sin(x) + (x^2) \cos(x) = 2x \sin(x) + x^2 \cos(x)$$

3.5 Quotient Rule

For $f(x) = \frac{u(x)}{v(x)}$, the derivative is:

$$f'(x) = \frac{u'(x) \cdot v(x) - u(x) \cdot v'(x)}{[v(x)]^2}$$

Example 12. Differentiate $f(x) = \frac{x^2}{\cos(x)}$.

Solution: Let $u(x) = x^2$ and $v(x) = \cos(x)$:

$$u'(x) = 2x, \quad v'(x) = -\sin(x)$$

$$\begin{aligned} f'(x) &= \frac{(2x \cdot \cos(x)) - (x^2 \cdot (-\sin(x)))}{\cos^2(x)} \\ &= \frac{2x \cos(x) + x^2 \sin(x)}{\cos^2(x)} \end{aligned}$$

3.6 Chain Rule

For $f(x) = g(h(x))$, the derivative is:

$$f'(x) = g'(h(x)) \cdot h'(x)$$

Example 13. Differentiate $f(x) = \sin(x^2)$.

Solution: Let $g(u) = \sin(u)$ and $h(x) = x^2$:

$$\begin{aligned}g'(u) &= \cos(u), & h'(x) &= 2x \\f'(x) &= \cos(x^2) \cdot 2x = 2x \cos(x^2)\end{aligned}$$

3.7 Techniques of Differentiation

3.8 Trigonometric Functions

$$\begin{aligned}\frac{d}{dx} \sin(x) &= \cos(x) \\ \frac{d}{dx} \cos(x) &= -\sin(x)\end{aligned}$$

Example 14. Differentiate $f(x) = \sin(x) \cdot \cos(x)$.

Solution: Using the product rule:

$$\begin{aligned}f'(x) &= \sin(x) \cdot (-\sin(x)) + \cos(x) \cdot \cos(x) \\ &= -\sin^2(x) + \cos^2(x)\end{aligned}$$

3.9 Exponential Functions

$$\begin{aligned}\frac{d}{dx} e^x &= e^x \\ \frac{d}{dx} a^x &= a^x \ln(a)\end{aligned}$$

Example 15. Differentiate $f(x) = e^{3x}$.

Solution: Using the chain rule:

$$f'(x) = e^{3x} \cdot 3 = 3e^{3x}$$

3.10 Logarithmic Functions

$$\begin{aligned}\frac{d}{dx} \ln(x) &= \frac{1}{x} \\ \frac{d}{dx} \log_a(x) &= \frac{1}{x \ln(a)}\end{aligned}$$

Example 16. Differentiate $f(x) = \ln(x^2 + 1)$.

Solution: Using the chain rule:

$$f'(x) = \frac{1}{x^2 + 1} \cdot 2x = \frac{2x}{x^2 + 1}$$

3.11 Polynomial Functions

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

Example 17. Differentiate $f(x) = 5x^4 - 3x^3 + x - 4$.

Solution:

$$f'(x) = 20x^3 - 9x^2 + 1$$

3.12 Implicit Functions

An implicit function is a relation defined by an equation involving two or more variables where one variable cannot be explicitly expressed in terms of the others.

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Example 18. Differentiate the implicit function defined by the equation:

$$x^2 + y^2 = 1$$

This represents a circle of radius 1 centered at the origin.

Solution: To differentiate this implicitly with respect to x , we will use implicit differentiation.

1. Differentiate both sides of the equation with respect to x :

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = \frac{d}{dx}(1)$$

This gives:

$$2x + 2y \frac{dy}{dx} = 0$$

2. Solve for $\frac{dy}{dx}$:

$$\begin{aligned} 2y \frac{dy}{dx} &= -2x \\ \frac{dy}{dx} &= -\frac{x}{y} \end{aligned}$$

4 Applications of Differentiation

4.1 Rates of Change

The derivative represents the rate of change of a function.

Example 19. *The position of an object is given by $s(t) = t^3 - 6t^2 + 9t$. Find the velocity at $t = 2$.*

Solution: Velocity is the derivative of the position function:

$$s'(t) = 3t^2 - 12t + 9$$

At $t = 2$:

$$s'(2) = 3(2^2) - 12(2) + 9 = 12 - 24 + 9 = -3$$

4.2 Optimization Problems

Optimization involves finding the maximum or minimum values of a function.

Example 20. *Example Find the maximum value of $f(x) = -x^2 + 4x - 3$.*

Solution: First, find the derivative:

$$f'(x) = -2x + 4$$

Set the derivative to zero to find critical points:

$$-2x + 4 = 0 \quad \Rightarrow \quad x = 2$$

Evaluate $f(x)$ at $x = 2$:

$$f(2) = -(2)^2 + 4(2) - 3 = -4 + 8 - 3 = 1$$

The maximum value is 1 .

You are making right steps, you are now going to look at integration.

5 Integration: Techniques and Application

5.1 Integration: Basic Concepts and Rule of n

Integration is the process of finding the antiderivative of a function, i.e., determining a function whose derivative is the given function. The integral of a function $f(x)$ with respect to x is denoted as:

$$\int f(x)dx$$

The Rule of n states that for any real number $n \neq -1$:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

where C is the constant of integration.

5.2 Techniques of Integration

5.3 Integration of Linear Functions

For linear functions, the integration is straightforward:

$$\int (ax + b) dx = \frac{a}{2}x^2 + bx + c$$

Example 21. Integrate $\int (3x + 4) dx$.

Solution

$$\int (3x + 4) dx = \frac{3}{2}x^2 + 4x + C$$

5.4 Integration by Substitution

Use substitution to simplify integrals where a function and its derivative appear.

Example 22. Integrate $\int 2xe^{x^2} dx$.

Solution: Let $u = x^2$, then $du = 2x dx$:

$$\int 2xe^{x^2} dx = \int e^u du = e^u + C = e^{x^2} + C$$

5.5 Integration of Trigonometric Powers

5.5.1 Even Powers

For even powers of sine or cosine, use identities to simplify.

Example 23. Integrate $\int \cos^2(x) dx$.

Solution: Use the identity $\cos^2(x) = \frac{1+\cos(2x)}{2}$:

$$\int \cos^2(x) dx = \int \frac{1 + \cos(2x)}{2} dx = \frac{x}{2} + \frac{1}{4} \sin(2x) + C$$

5.5.2 Odd Powers

For odd powers, use trigonometric identities and substitution.

Example 24. Integrate $\int \sin^3(x) dx$.

Solution: Use $\sin^3(x) = \sin(x)(1 - \cos^2(x))$:

$$\int \sin^3(x) dx = \int \sin(x) - \sin(x) \cos^2(x) dx$$

Let $u = \cos(x)$, then $du = -\sin(x)dx$:

$$\begin{aligned}\int \sin(x)dx - \int \sin(x) \cos^2(x)dx &= -\cos(x) - \int (1 - u^2) (-du) \\ &= -\cos(x) + u - \frac{u^3}{3} + C = -\cos(x) + \cos(x) - \frac{\cos^3(x)}{3} + C\end{aligned}$$

5.6 Integration Using Reduction Formulae

Reduction formulae are used to simplify the integration of powers of trigonometric functions.

Example 25. Integrate $\int \sin^4(x)dx$.

Solution: Use the reduction formula:

$$\begin{aligned}\int \sin^4(x)dx &= \frac{3}{4} \int \sin^2(x)dx - \frac{1}{4} \int \sin^0(x)dx \\ \text{Using } \int \sin^2(x)dx &= \frac{x}{2} - \frac{1}{4} \sin(2x) : \\ \int \sin^4(x)dx &= \frac{3}{4} \left(\frac{x}{2} - \frac{1}{4} \sin(2x) \right) - \frac{x}{4} \\ &= \frac{3x}{8} - \frac{3}{16} \sin(2x) - \frac{x}{4} = \frac{3x}{8} - \frac{x}{4} - \frac{3}{16} \sin(2x)\end{aligned}$$

5.7 Integration of Products of Trigonometric Functions

Use trigonometric identities to simplify products.

Example 26. Integrate $\int \sin(x) \cos(x)dx$.

Solution: Use the identity $\sin(x) \cos(x) = \frac{1}{2} \sin(2x)$:

$$\begin{aligned}\int \sin(x) \cos(x)dx &= \frac{1}{2} \int \sin(2x)dx \\ &= -\frac{1}{4} \cos(2x) + C\end{aligned}$$

5.8 Integration of Logarithmic and Exponential Functions

Integration of Logarithmic Functions:

$$\int \ln(x)dx = x \ln(x) - x + C$$

Integration of Exponential Functions:

$$\begin{aligned}\int e^x dx &= e^x + C \\ \int a^x dx &= \frac{a^x}{\ln(a)} + C\end{aligned}$$

Example 27. Integrate $\int xe^x dx$.

Solution: Use integration by parts:

$$\int xe^x dx = xe^x - \int e^x dx = xe^x - e^x + C$$

5.9 Integration by Parts

$$\int u dv = uv - \int v du$$

Example 28. Integrate $\int x \ln(x) dx$.

Solution: Let $u = \ln(x)$ and $dv = x dx$:

$$\begin{aligned} du &= \frac{1}{x} dx \text{ and } v = \frac{x^2}{2} \\ \int x \ln(x) dx &= \frac{x^2}{2} \ln(x) - \int \frac{x^2}{2} \cdot \frac{1}{x} dx \\ &= \frac{x^2}{2} \ln(x) - \int \frac{x}{2} dx = \frac{x^2}{2} \ln(x) - \frac{x^2}{4} + C \end{aligned}$$

5.10 Partial Fractions

Decompose rational functions into simpler fractions.

Example 29. Integrate $\int \frac{1}{x^2-1} dx$.

Solution: Decompose:

$$\begin{aligned} \frac{1}{x^2-1} &= \frac{1}{(x-1)(x+1)} \\ \frac{1}{x^2-1} &= \frac{A}{x-1} + \frac{B}{x+1} \end{aligned}$$

Solve for A and B :

$$1 = A(x+1) + B(x-1)$$

Setting $x = 1$ and $x = -1$:

$$\begin{aligned} A &= \frac{1}{2}, \quad B = -\frac{1}{2} \\ \int \frac{1}{x^2-1} dx &= \frac{1}{2} \int \frac{1}{x-1} dx - \frac{1}{2} \int \frac{1}{x+1} dx \\ &= \frac{1}{2} \ln|x-1| - \frac{1}{2} \ln|x+1| + C \\ &= \frac{1}{2} \ln \left| \frac{x-1}{x+1} \right| + C \end{aligned}$$

5.11 Integration Using Inverse Trigonometric Substitution

Use inverse trigonometric functions to simplify integrals involving quadratic expressions.

Example 30. Integrate $\int \frac{1}{\sqrt{1-x^2}} dx$.

Solution: Use substitution $x = \sin(\theta)$:

$$\begin{aligned} dx &= \cos(\theta) d\theta \\ \int \frac{1}{\sqrt{1-\sin^2(\theta)}} \cos(\theta) d\theta &= \int d\theta = \theta + C \\ \theta = \arcsin(x) \text{ so } \int \frac{1}{\sqrt{1-x^2}} dx &= \arcsin(x) + C \end{aligned}$$

6 Applications of Integration

6.1 Area Under a Curve

Example 31. Find the area under $f(x) = x^2$ from $x = 0$ to $x = 2$.

Solution:

$$\text{Area} = \int_0^2 x^2 dx = \left[\frac{x^3}{3} \right]_0^2 = \frac{2^3}{3} - \frac{0^3}{3} = \frac{8}{3}$$

6.2 Volume of Revolution

Example 32. Find the volume of the solid formed by rotating $f(x) = x^2$ about the x -axis from $x = 0$ to $x = 2$.

Solution:

$$\begin{aligned} \text{Volume} &= \pi \int_0^2 (x^2)^2 dx = \pi \int_0^2 x^4 dx \\ &= \pi \left[\frac{x^5}{5} \right]_0^2 = \pi \left(\frac{2^5}{5} \right) = \frac{32\pi}{5} \end{aligned}$$

6.3 Motion: Distance Traveled

Example 33. The velocity of an object is given by $v(t) = 3t^2$. Find the distance traveled from $t = 1$ to $t = 3$.

Solution:

$$\begin{aligned} \text{Distance} &= \int_1^3 3t^2 dt = [t^3]_1^3 \\ &= 3^3 - 1^3 = 27 - 1 = 26 \end{aligned}$$

6.4 Applications in AI Problems

Example 34. In a machine learning model, the loss function is defined as:

$$L(\theta) = \int_0^1 \theta^2 \sin(\pi x) dx$$

Find the value of θ that minimizes the loss function $L(\theta)$.

Solution: We calculate the integral for the loss function $L(\theta)$:

$$L(\theta) = \int_0^1 \theta^2 \sin(\pi x) dx = \theta^2 \int_0^1 \sin(\pi x) dx$$

The integral of $\sin(\pi x)$ over $[0, 1]$ is $\frac{2}{\pi}$, so:

$$L(\theta) = \theta^2 \frac{2}{\pi}$$

To minimize $L(\theta)$, we solve $\frac{dL(\theta)}{d\theta} = 0$, which gives $\theta = 0$.

Interpretation: In machine learning, the loss function $L(\theta)$ measures the discrepancy between the predicted outputs and the actual target values given model parameters θ . To understand the performance of a model at a specific parameter value, such as $\theta = 0$, we evaluate $L(\theta)$ at this point. The interpretation of $L(\theta)$ at $\theta = 0$ provides insight into the model's performance with zero parameters. It serves as a baseline measure indicating the extent of work required to adjust θ for improved performance. Understanding this baseline helps in evaluating parameter tuning and overall model improvement.

7 Gradient Descent in Optimization

Consider a simple function

$$f(x) = x^3 - 3x^2 + 4$$

that represents the error of a machine learning model.

- Compute the first derivative $f'(x)$ and identify the critical points.
- Using a numerical approximation, illustrate how gradient descent would work starting from an initial point $x_0 = 0$. Assume a learning rate $\alpha = 0.1$.
- Discuss how the concepts of derivatives and gradients are used in training AI models.